Sai Sriram Duddu

Genova, Italy

LinkedIn
Git
+39 3446458545
sriram.sai98@gmail.com

Education

EIT Manufacturing Double Degree in Robotics: Mondragon University, Basque, Spain and TU Wien, Austria.

Sept 2022-August 2025

- Year 1 **Mondragon University**: Gained comprehensive insights into Robotics and Control, acquiring a broad understanding and project-based learning experience across various fields. This includes Mechanics, Electronic Sensors and Actuators, their control systems, Signal Processing, Programming and Simulation, Communication Protocols, Machine Vision, and in-depth aspects of Machine Learning, including Deep Learning and Reinforcement Learning.
- Year 2 **TUWien**: Delved deeper into research-oriented topics, such as writing literature reviews and presenting papers on a wide range of current trends including Explainable AI (XAI), Human-Robot Interaction, Intelligent User Agents, Robotic Additive Manufacturing, and Haptic Technologies. Additionally, implemented numerous use-case scenarios in Assistance Systems, focusing on Collaborative Robots and Passive Exoskeletons.

Bachelor of Engineering in Mechanical Engineering: Andhra University, India June 2015-July 2019

- I gained a profound understanding of concepts in physics and mathematics, alongside a robust foundation in mechanics, machine design, and mechatronics. I also acquired in-depth practical knowledge in workshop technologies and manufacturing, which later played a significant role in my comprehension of robot modelling and building. Additionally, during this period, I had the opportunity to participate in numerous national robotics competitions and attend insightful conferences.

Work Experience

Student Research Fellow ,HHCM Lab - Istituto Italiano de Tecnologia : Genova, Italy Sep 2024-July 2025

- Involved in an Active European Project called "Magician" developing a Novel passive complient based end effector interface for a Robotic arm that could work collaboratively in the Automotive Industry.
- Being actively involved in both the design as well as software aspects of the above-mentioned project.

GRADUATE APPRENTICE TRAINEE, Vizag Steel Plant: Vizag, India Jan 2021-Jan 2022

- Inspected and troubleshot various machinery and mechanisms essential to steel manufacturing, such as gearboxes, couplings, and hydraulic industrial machines.
- Played a key role in testing the ABB IRB 140 robotic arm for automated sample testing to analyze the metallurgical properties of steel.

QUANTUM COMPUTING INTERN, IBM: Online, Remote Oct 2020-May 2021

- This internship provided an overview of quantum mechanics fundamentals and their applications in computation. I gained insights into key algorithms, such as Shor's algorithm for factoring and Grover's algorithm for search, and learned how quantum circuits (qubits) function.
- Explored various quantum computer hardware. Towards the end of this program, I had the privilege of implementing some arithmetic algorithms using qubits on an actual "IBM Q 53" through a remote server.

CAD AND 3D PRINTING DESIGNER, INDIA ASTRONOMY CLUB: Vizag, India Jan 2020-Jan 2021

- Designed and prototyped optical and mechanical components for Mechanized telescopic mounts.
- Crafted a custom 8" Dobsonian telescope with 3D printed parts. By doing so, we were able to provide this custom telescope to our clients for a third of the actual cost. Furthermore, our design is much more modular and easier to transport.

Projects

- **Currently Involved in One of the Active European Projects called "Magician" -An Immersive learning for imperfection detection and repair through human-robot interaction: Link.**
- **Envolved in a Project of making a "F1tenth Autonomous Racing Car".**
- **Modelled and simulated a UR-3 Cobot in Unity for the Use Case of XAI.**
- **Developed a Novel approach for Packing in Warehouses using Cobot.**
- **Reconstruction of Objects in a 3d Point cloud using Robotic arm Stereo vision.**
- **Development of Autonomous Mobile Robot Using Reinforcement Learning (Q-learning and DDQN).**
- **Generated a Kalman filter algorithm for parabolic projects and enhanced it by reducing the noise.**
- **Deep Learning in Weld Analysis by training the ML Model to detect the Weld defects.**
- **Line follower and obstacle detector using a TurtleBot.**
- **Automated Robotic Screw Inspection Equipment by using Artificial Vision Algorithms.**
- **Omnidirectional Agile Robot for Industrial and Autonomous Vehicle Applications.**
- **Design and Development of a 3D-Printed Delta Robot Controlled by Inverse Kinematics.**
- **3D-Printed Arduino-Controlled Desktop Robotic Arm with 4 Degrees of Freedom.**

Skills

- Programming Languages: C/C++, C sharp, Python, Embedded C.
- Tools: Siemens NX, Creo Parametric, SolidWorks, Fusion360, MathWorks, Unity, Ansys, Git, Linux, Microsoft tools, Latex, Docker.
- Frameworks: ROS1 and Ros2, OpenCV, Gazebo, RobotStudio-ABB, PolyScope, Franka-interface.
- Machine Learning: TensorFlow, Keras, PyTorch, Scikit, Qiskit.
- Communication: English(Fluent),Hindi(Fluent).

Publications

- Ramya, P., & Sai Sriram,D. (2019). Design and fabrication of model jet engine using recycled turbocharger parts. *International Journal for Research in Applied Science and Engineering Technology*, 7(4), 606–613. doi:10.22214/ijraset.2019.4110

Recent Honours and Awards

- Circular Dev Summer School -3rd Prize Winning Team.
- A Full Scholarship for Master's Program – European Institute of Technology (EIT) Manufacturing Master School.
- Won a Cash Prize worth 3000 euros for the best-making of Go kart that was raced in a national racing competition held by the Federation of Motor Sports Club of India.

Volunteering

- Service and Medical Volunteer.
- Robotics Students Volunteer.